

VIJU SANJAI

+91 7338944392 | avijusanjai@gmail.com | github.com/pocopepe | linkedin.com/in/vijusanjai

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication (GPA: 8.16)

Chennai, TN

Aug 2023 – Present

TECHNICAL SKILLS

Languages: Java, C/C++, C#, Python, Rust, SQL, JavaScript/TypeScript

Frameworks & Libraries: .NET, Express.js, Node.js, React, Tauri, WebGPU, PyTorch

Cloud & Tools: AWS, Cloudflare (R2, Edge), Docker, Git, GitHub Actions, Linux, Supabase, Vercel

Concepts/Areas: Data Structures & Algorithms, Object-Oriented Design, API Architecture, System Design, Unit Testing, CI/CD, Debugging

EXPERIENCE

Lagoon Technologies

Software Development Intern

Chennai, TN

Sep 2024 – Feb 2025

- Engineered and optimized internal enterprise software tools utilizing **C# and the .NET framework** within an Agile environment.
- Conducted exploratory analysis of legacy systems, identifying and debugging structural bottlenecks to improve maintainability and accelerate release cycles.
- Participated in daily Agile standups and cross-functional code reviews to align software architecture with enterprise scalability requirements.

TECHNICAL PROJECTS

Share (Scalable Cloud Storage Service) | TypeScript, Hono, Cloudflare R2

Oct 2024 – Present

- Architected a scalable, real-time file-sharing platform sustaining **4,700+ req/sec at 21ms avg latency** under load; in-browser previews for PDFs, text, and video codecs with zero plugin dependency.
- Built a globally distributed storage backend on **Cloudflare R2** with edge-local retrieval; routing APIs via **Hono** and **TypeScript** deployed on Cloudflare Workers for minimal cold-start latency.
- Designed a clean RESTful API surface with typed request/response contracts end-to-end in TypeScript, enabling reliable client integration and straightforward extension.

FFmpeg WebGPU Integration | Rust, WebGPU, WGSL, C, FFmpeg

Dec 2024 – Present

- Engineered a custom **hwcontext** for FFmpeg, enabling efficient GPU buffer management and zero-copy hardware-accelerated video pipelines.
- Ported complex Vulkan-based video processing filters into **WGSL** compute shaders for cross-platform execution.
- Leveraged **Rust** to safely interface with FFmpeg's legacy C-based architecture, heavily optimizing memory ownership and reducing CPU latency.

Ed25519 Firmware Signer | Rust, ed25519-dalek, Argon2id, AES-256-GCM, Tauri

Jun 2024 – Present

- Built a secure desktop utility for signing firmware binaries; private key encrypted at rest via **Argon2id** KDF + **AES-256-GCM** — keys never touch disk in plaintext.
- Applied **zeroize** for guaranteed key material erasure from memory and **OsRng** for CSPRNG-seeded generation; integrated **libgit2** for auditable, git-tracked signing history.
- Exposed the Rust backend via a **Tauri** desktop UI using IPC, keeping the full application memory footprint minimal.

LEADERSHIP & COMMUNITY

- Code Chef Developer:** Built and maintained the club's competitive programming challenge platform, utilizing **Express.js** and **Cloudflare** to handle scalable backend service requests.
- OSPC Research Lead:** Mentored 50+ students on **Git** workflows, code review practices, unit testing fundamentals, and collaborative open-source development.